

Demographic Forecasting

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Harvard University

Joint work with Federico Girosi (RAND)
with contributions from Kevin Quinn and Gregory Wawro

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- Approach: Formalizing **qualitative** insights in **quantitative** models

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- Better ways of creating Bayesian priors
- Produces forecasts and farcasts using considerably more information

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	<u>% Improvement</u>	
	Over Best Previous	to Best Conceivable
Cardiovascular	22	49
Lung Cancer	24	47
Transportation	16	31
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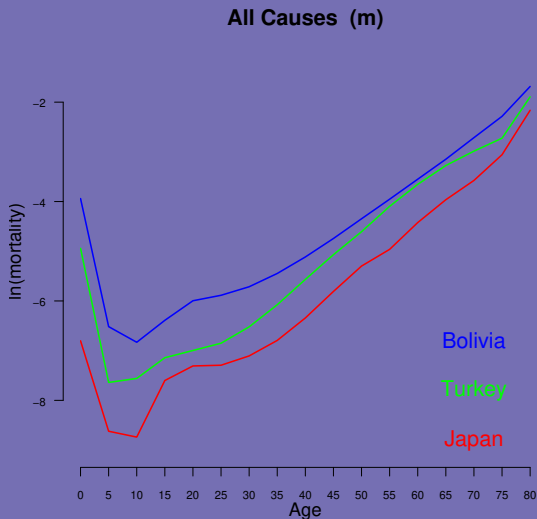
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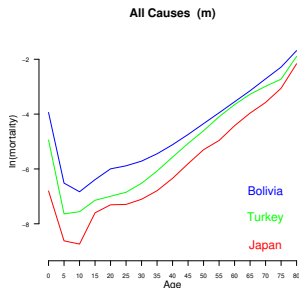
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- Does *considerably* better with **more informative covariates**

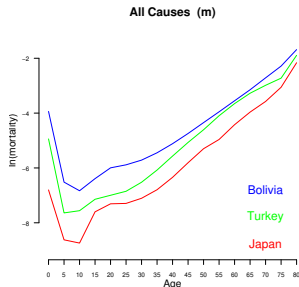
All-Cause Mortality Age Profile Patterns



Existing Method 1: Parameterize the Age Profile

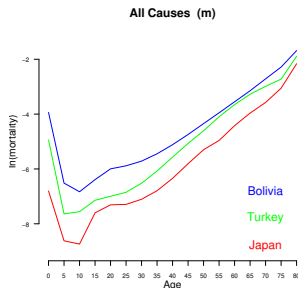


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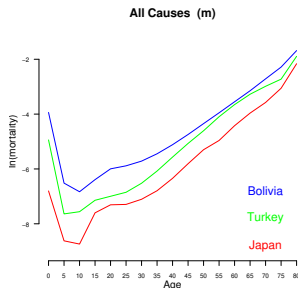
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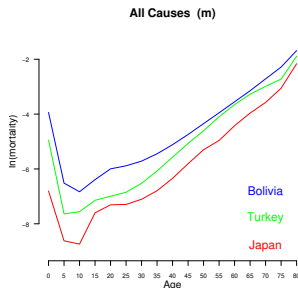
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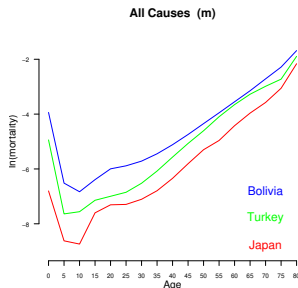
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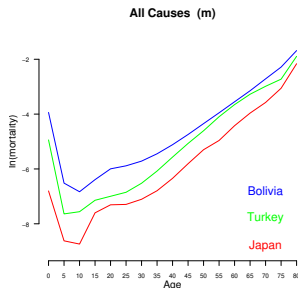
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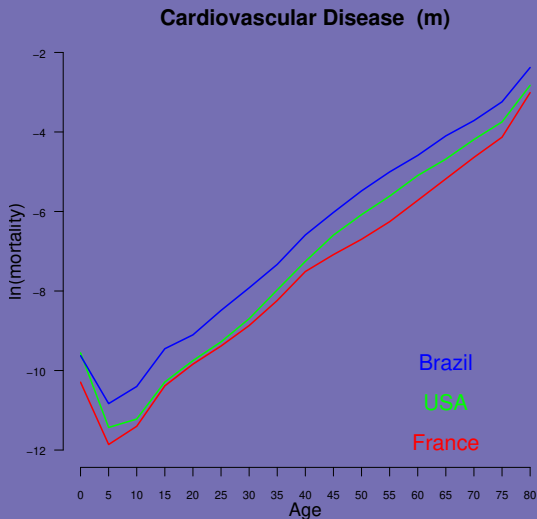
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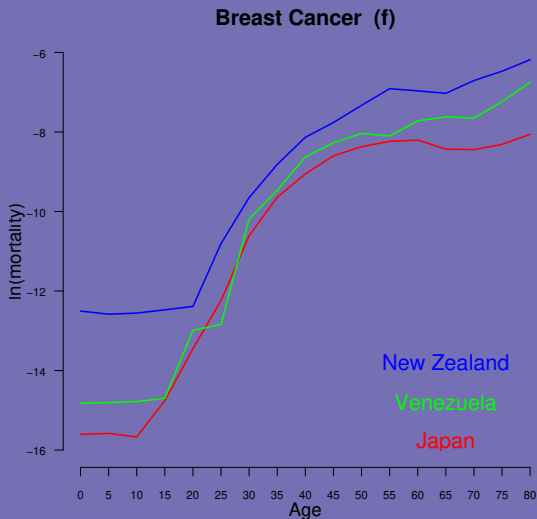


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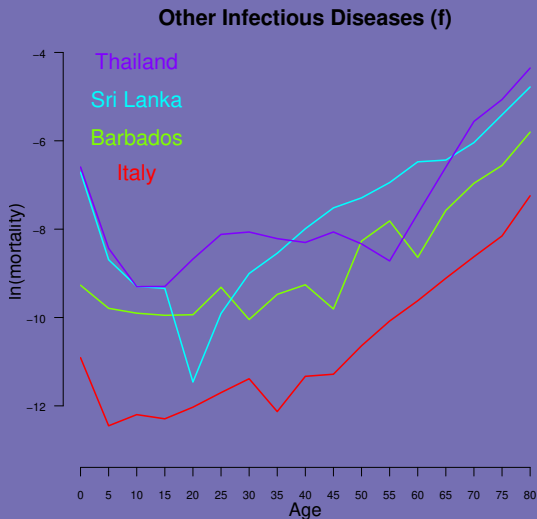
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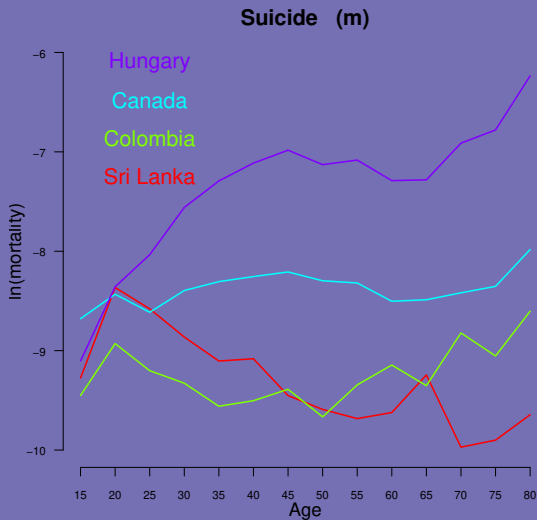
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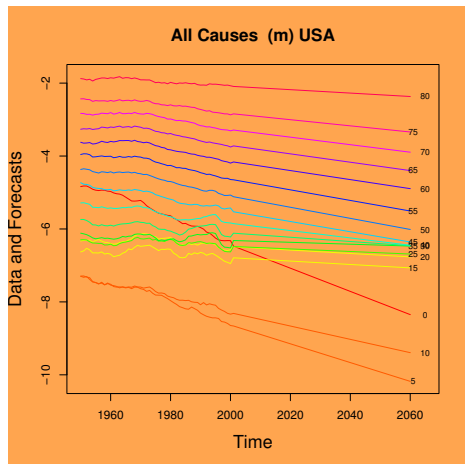
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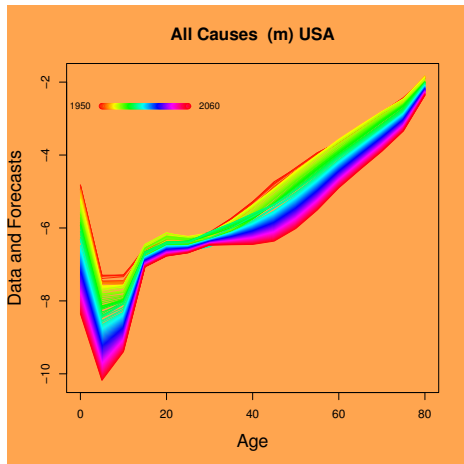
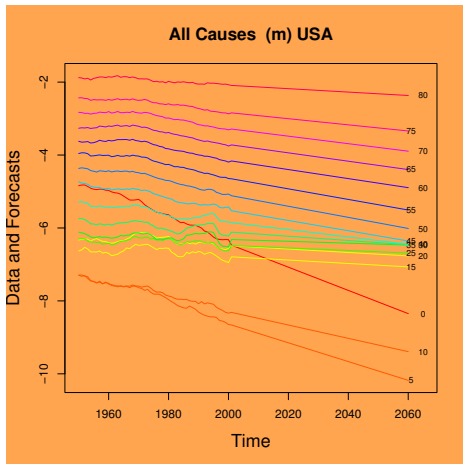
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- Also: Method ignores covariate information; the leading current method (McNown-Rogers) not replicable

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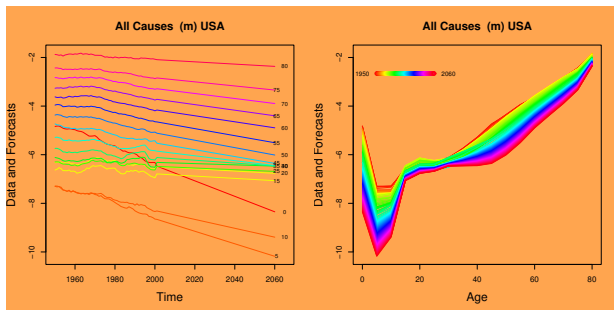
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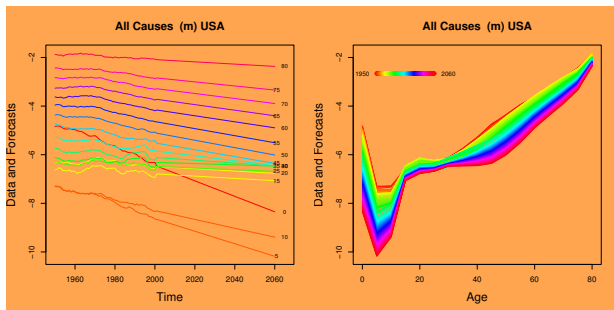
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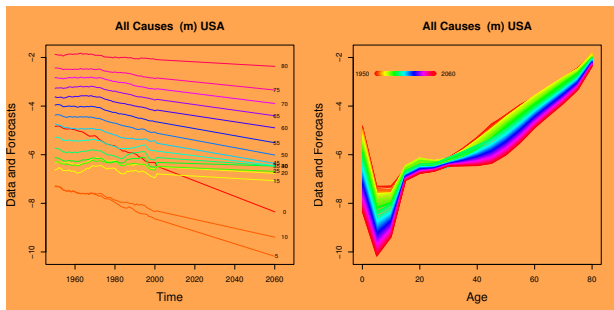


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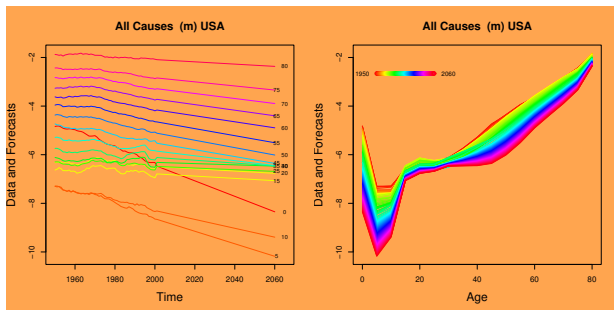
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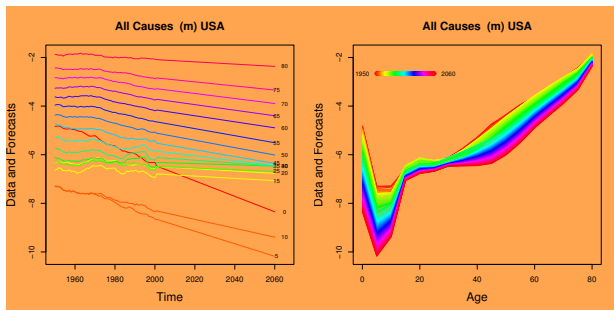
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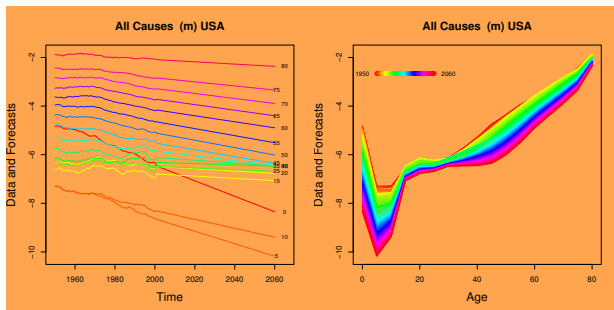
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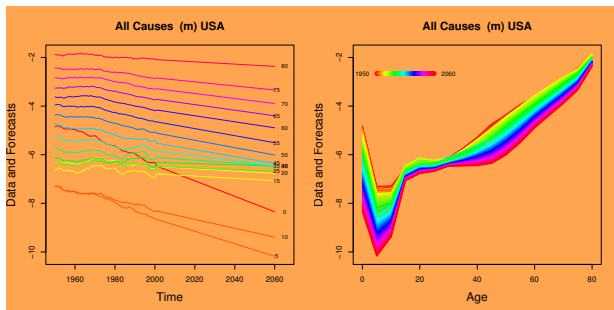
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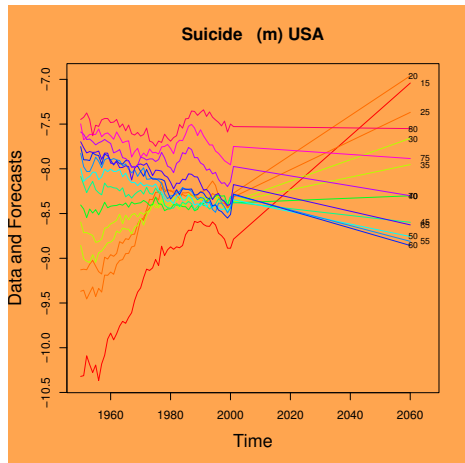
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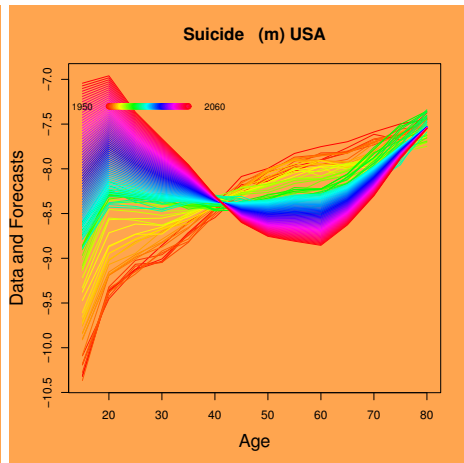
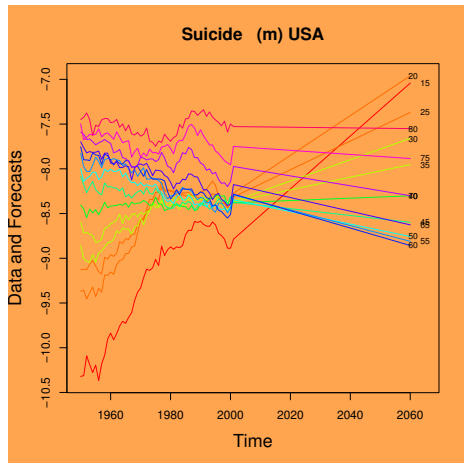
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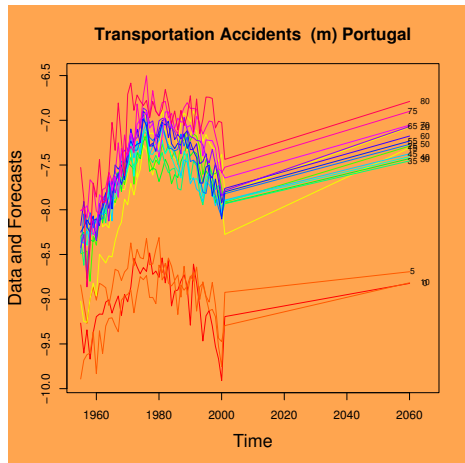
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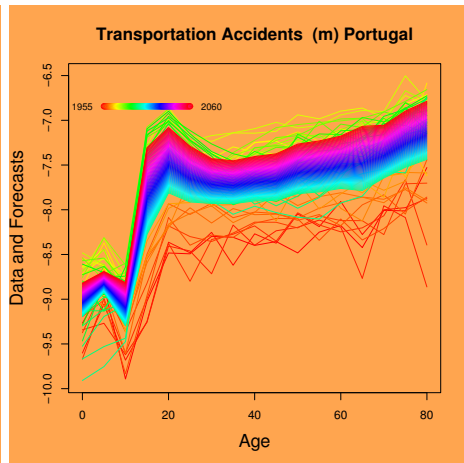
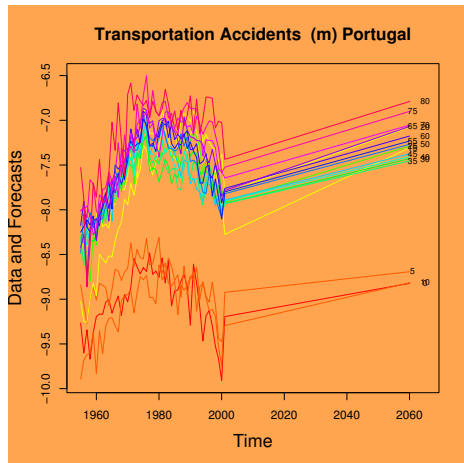
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Partial Pooling via a Bayesian Hierarchical Approach

- Likelihood for equation-by-equation least squares:

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- The easy part: *easy-to-use software* to implement everything we discuss today.

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Natural choice for the prior:

$$\mathcal{P}(\beta \mid \Phi) \propto \exp \left(-\frac{1}{2} \sum_{ij} s_{ij} \|\beta_i - \beta_j\|_{\Phi}^2 \right)$$

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- Extensive trial-and-error runs, yielded no useful parameter values.

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 - 3 In the subspace, we can invert $\mu = \mathbf{Z}\beta$ as $\beta = (\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mu$, giving:

$$\mathcal{P}(\beta \mid \theta) \propto \exp\left(-\frac{1}{2}H[\mu, \theta]\right) = \exp\left(-\frac{1}{2}H[\mathbf{Z}\beta, \theta]\right)$$

the same prior on μ , expressed as a function of β (with constant Jacobian).

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- We start with one-dimensional $P(\mu_{cat})$, and treat it as the multidimensional $P(\beta_{ca})$, constant in all directions but $Z_{cat}\beta_{ca}$.

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- Priors are based on knowledge rather than guesses.

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- where W^n is a matrix uniquely determined by n and θ

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where we have defined:

$$\mathbf{C}_{aa'} \equiv \frac{1}{T} \mathbf{Z}_a' \mathbf{Z}_{a'} \quad \mathbf{Z}_a \text{ is a } T \times d_a \text{ data matrix for age group } a$$

The Prior on the Coefficients β

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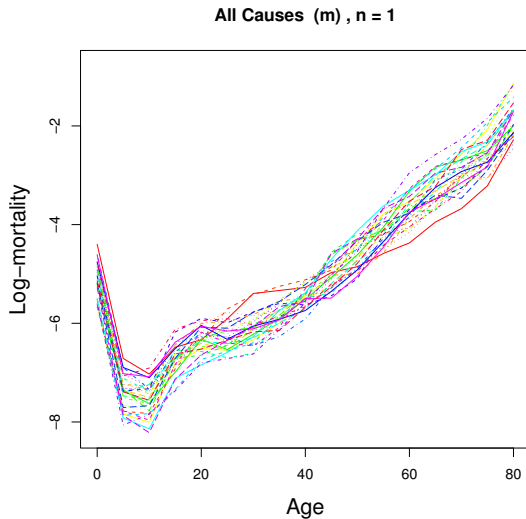
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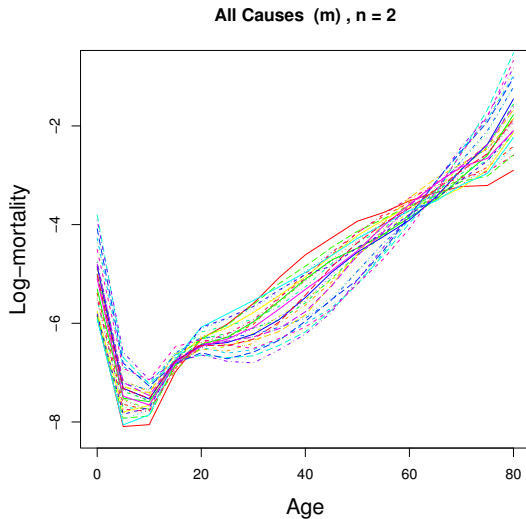
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- The variance of the prior is inversely proportional to θ , which controls the “strength” of the prior.

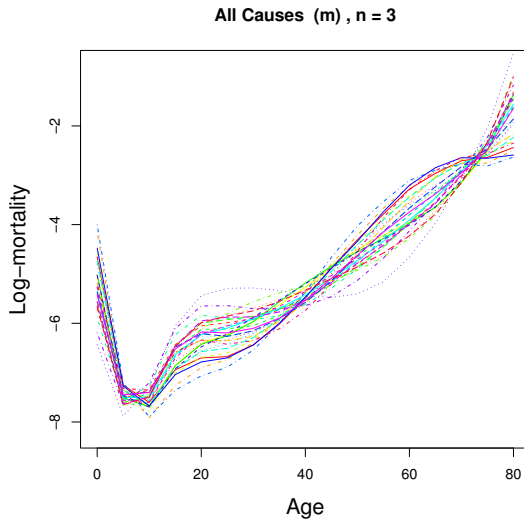
Samples From Age Prior



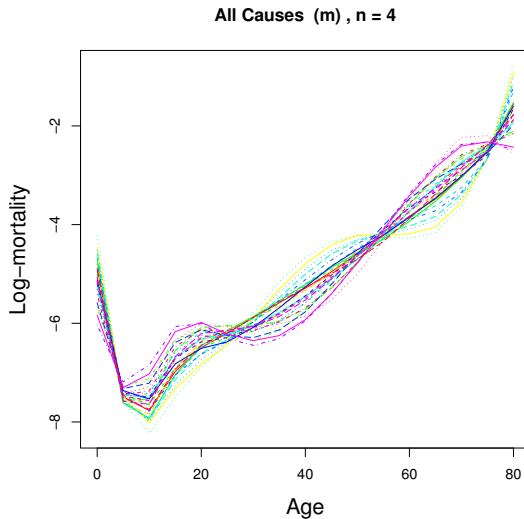
Samples From Age Prior



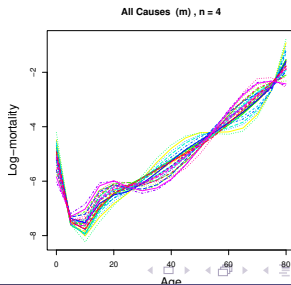
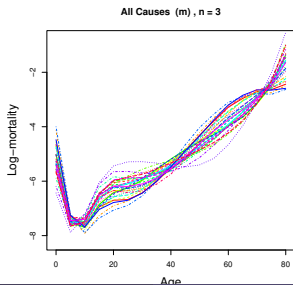
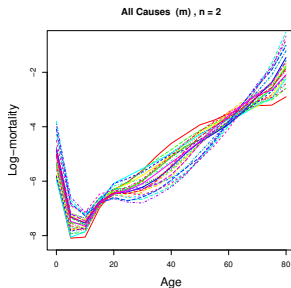
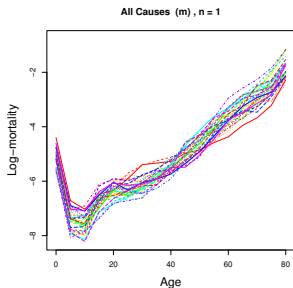
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Prior Indifference

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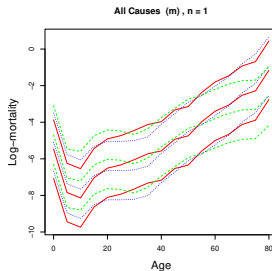
Formalizing (Prior) Indifference

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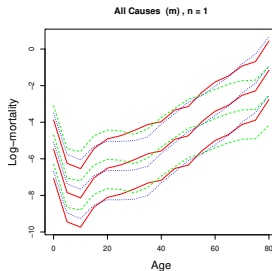
Level indifference



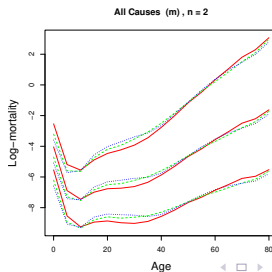
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Level and slope indifference



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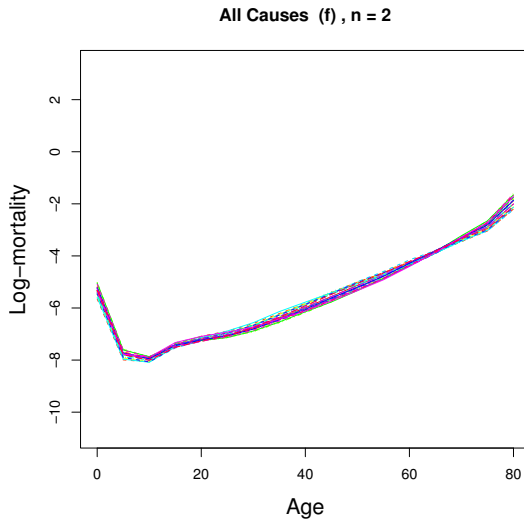
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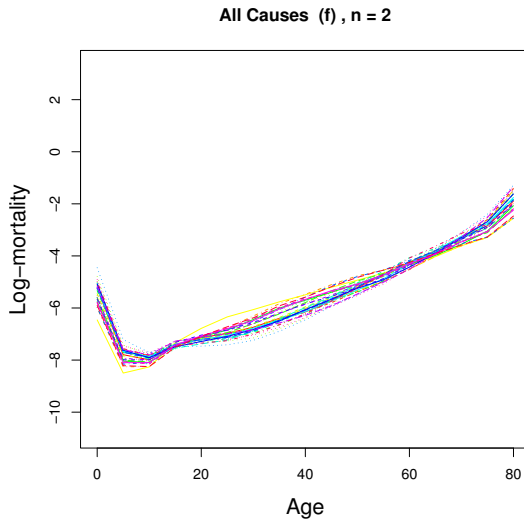
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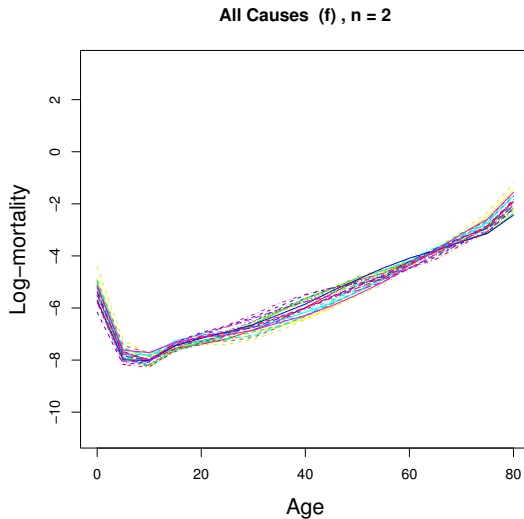
Samples from Age Prior



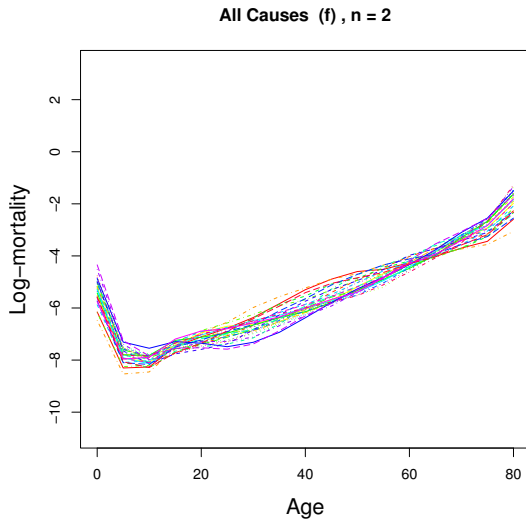
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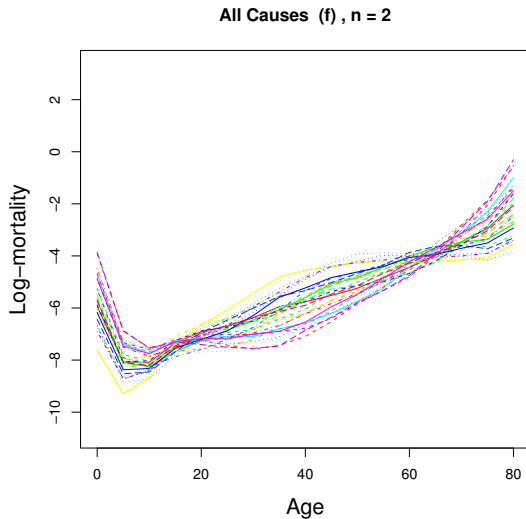
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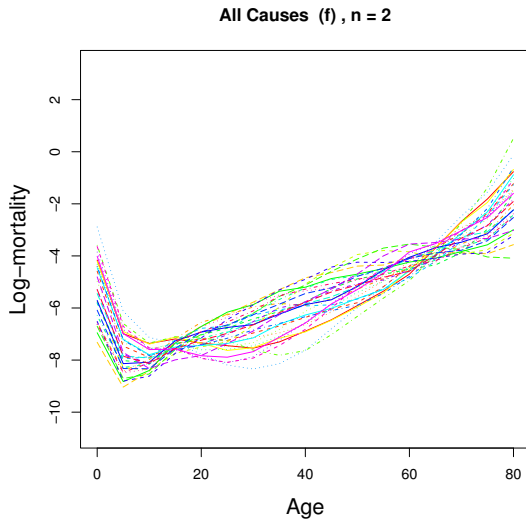
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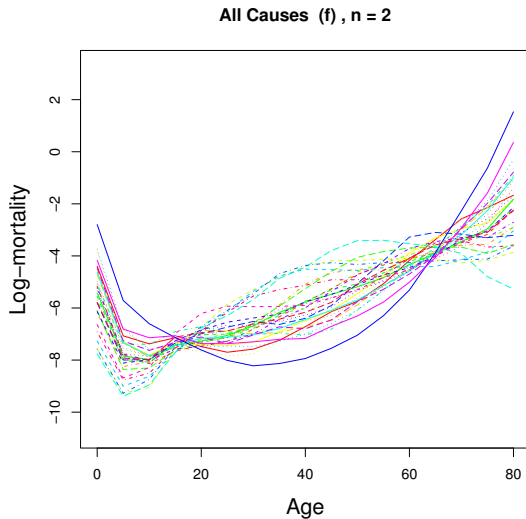
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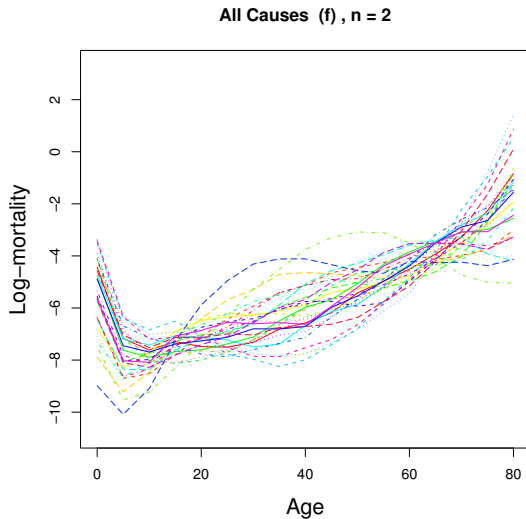
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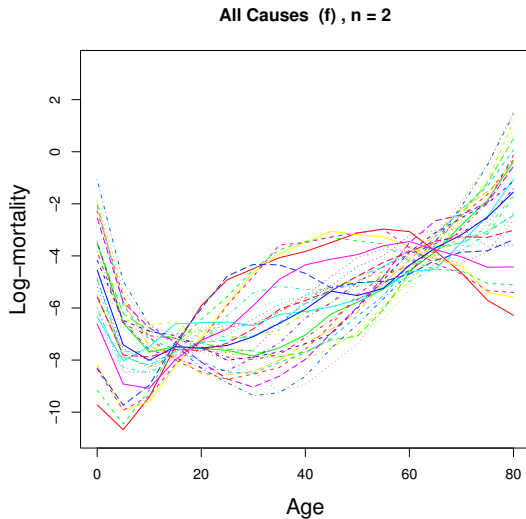
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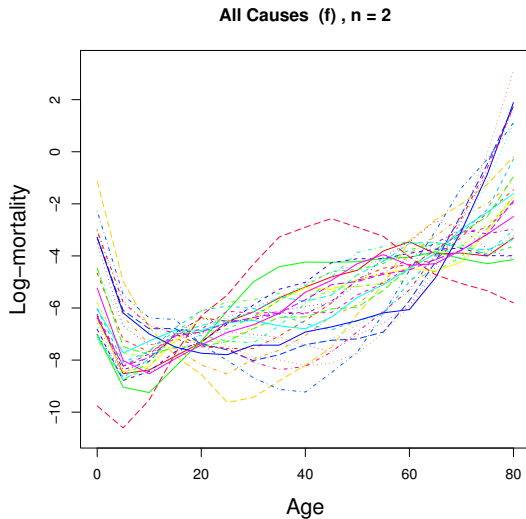
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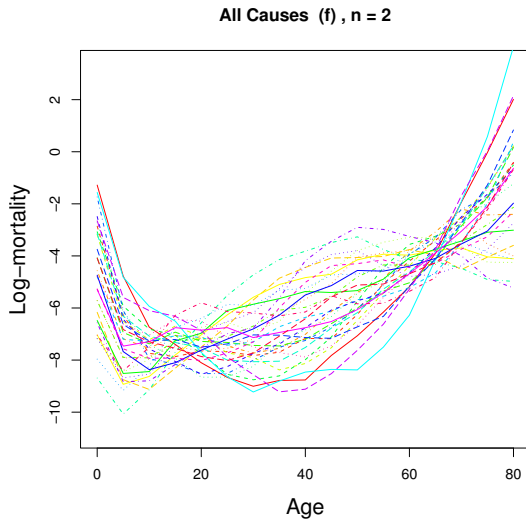
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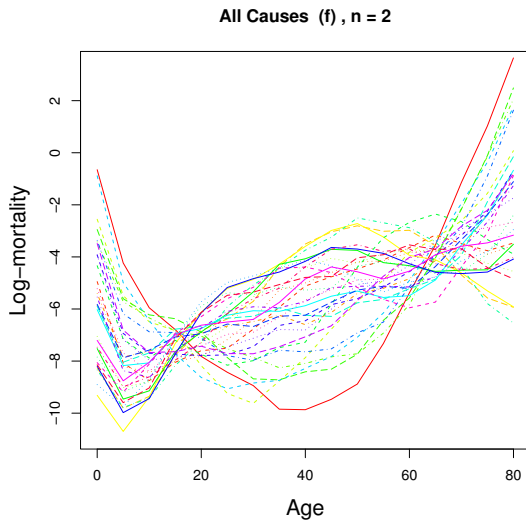
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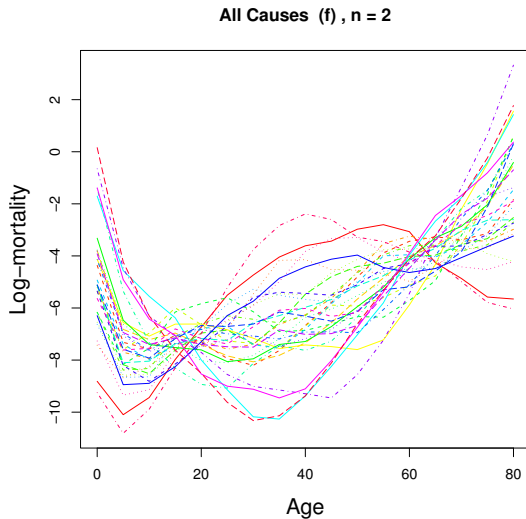
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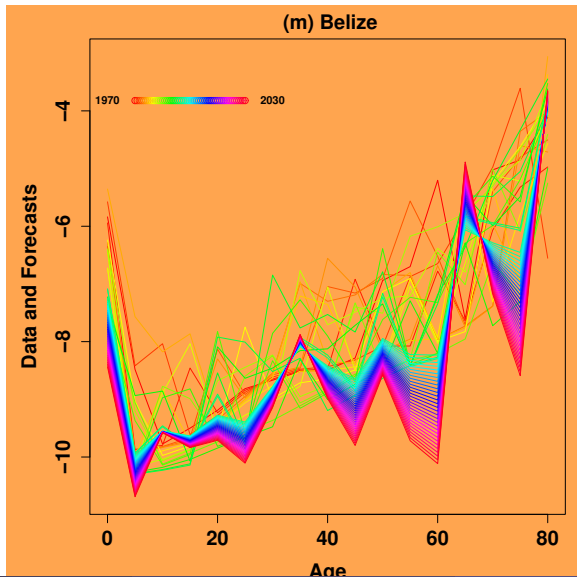
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- The mathematical form for *all* these (separately or together) turns out to be the same:

$$\mathcal{P}(\boldsymbol{\beta} \mid \theta) \propto \exp \left(-\frac{\theta}{2} \sum_{ij} W_{ij} \boldsymbol{\beta}'_i \mathbf{C}_{ij} \boldsymbol{\beta}_j \right), \quad \mathbf{C}_{aa'} \equiv \frac{1}{T} \mathbf{Z}_a \mathbf{Z}_{a'}$$

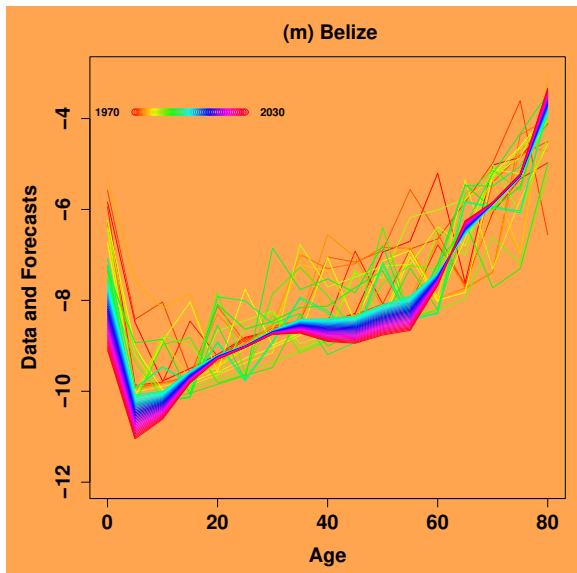
Mortality from Respiratory Infections, Males

Least Squares



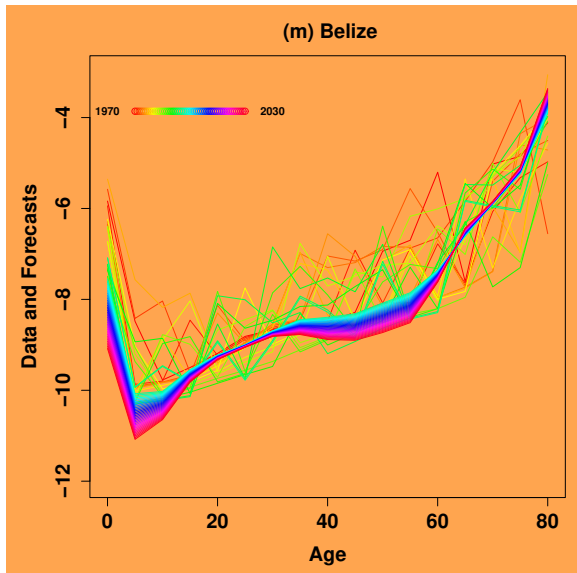
Mortality from Respiratory Infections, males, $\sigma = 2.00$

Smoothing over Age Groups



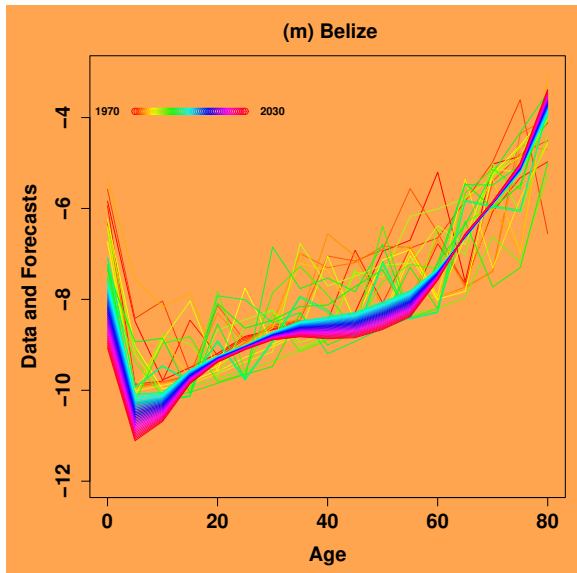
Mortality from Respiratory Infections, males, $\sigma = 1.51$

Smoothing over Age Groups



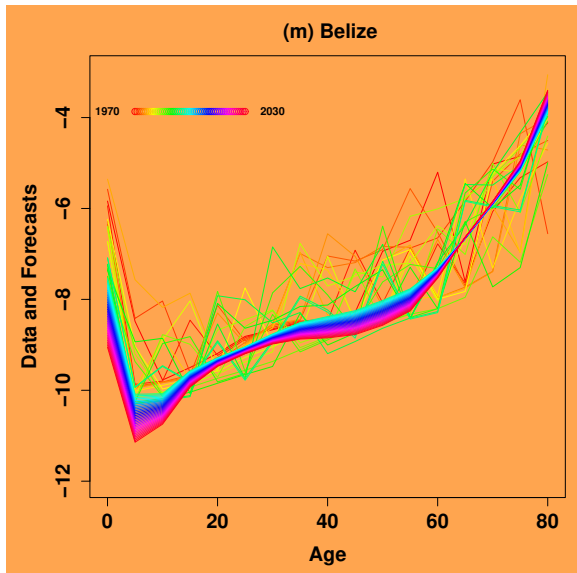
Mortality from Respiratory Infections, males, $\sigma = 1.15$

Smoothing over Age Groups



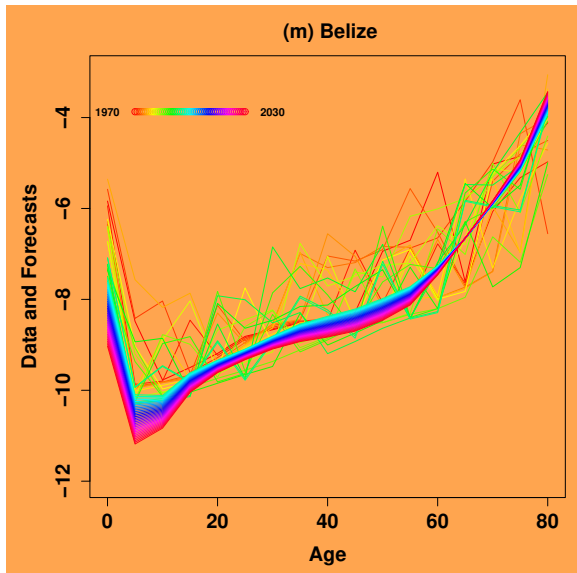
Mortality from Respiratory Infections, males, $\sigma = 0.87$

Smoothing over Age Groups



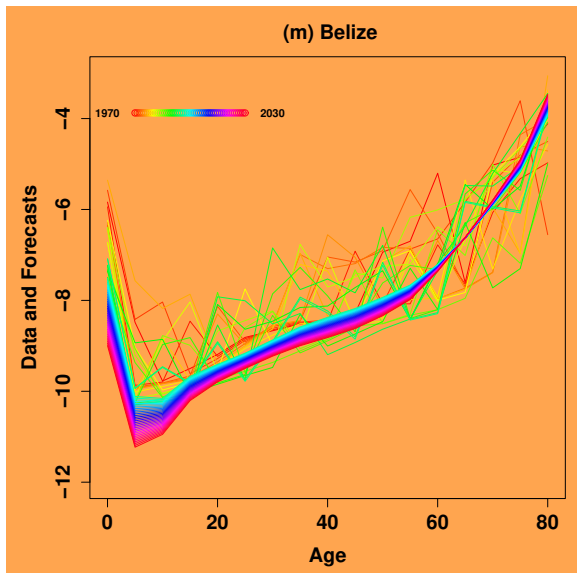
Mortality from Respiratory Infections, males, $\sigma = 0.66$

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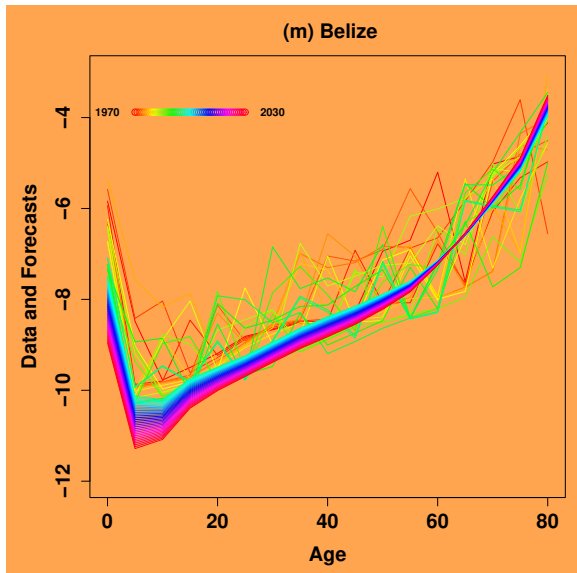
Mortality from Respiratory Infections, males, $\sigma = 0.50$

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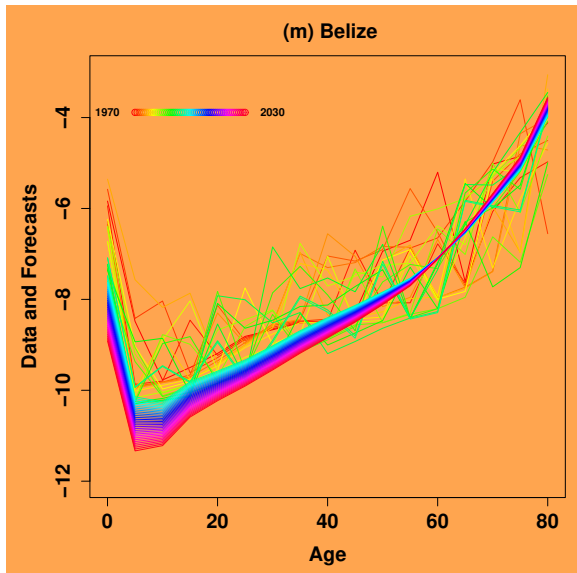
Mortality from Respiratory Infections, males, $\sigma = 0.38$

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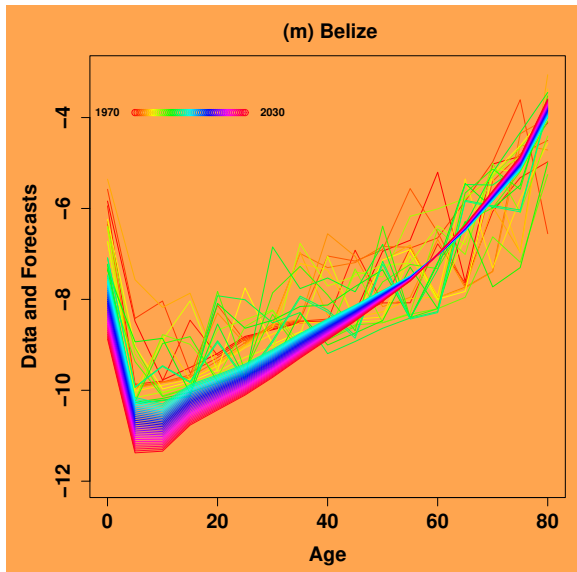
Mortality from Respiratory Infections, males, $\sigma = 0.28$

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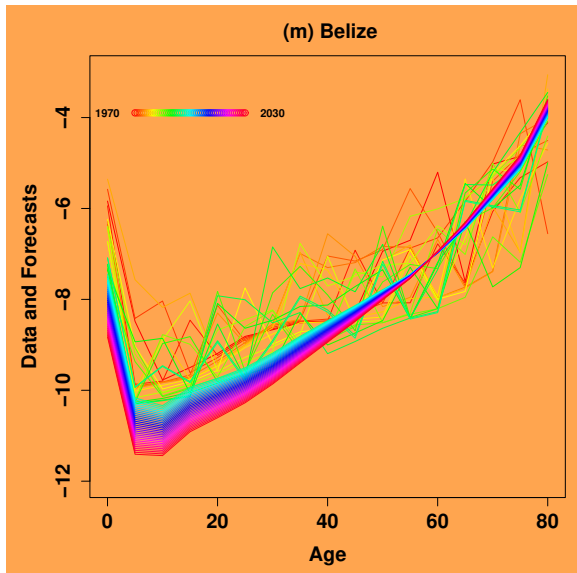
Mortality from Respiratory Infections, males, $\sigma = 0.21$

Smoothing over Age Groups



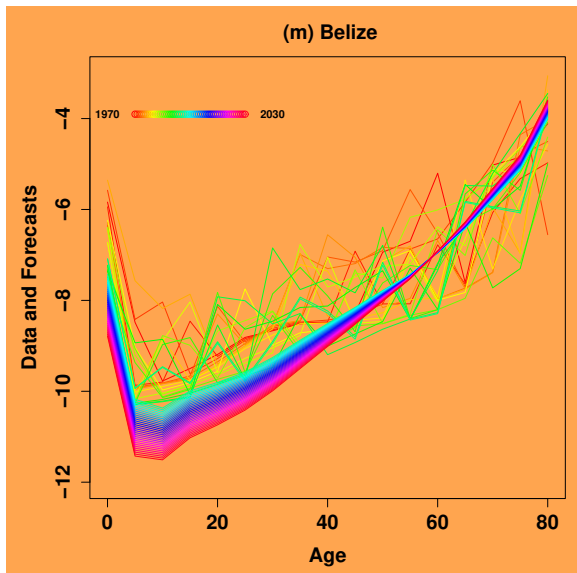
Mortality from Respiratory Infections, males, $\sigma = 0.16$

Smoothing over Age Groups



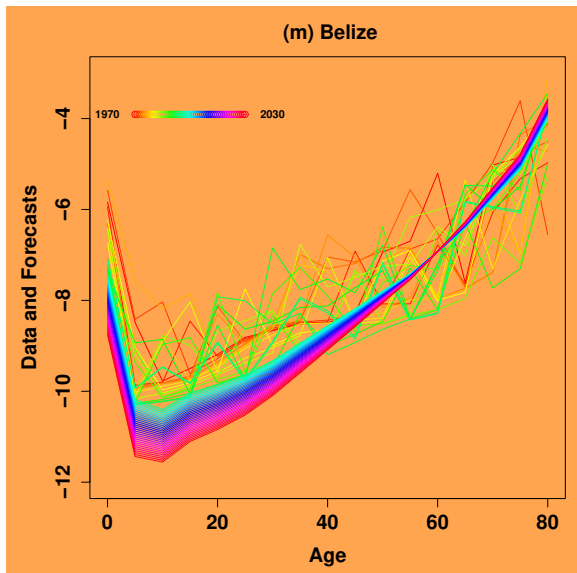
Mortality from Respiratory Infections, males, $\sigma = 0.12$

Smoothing over Age Groups



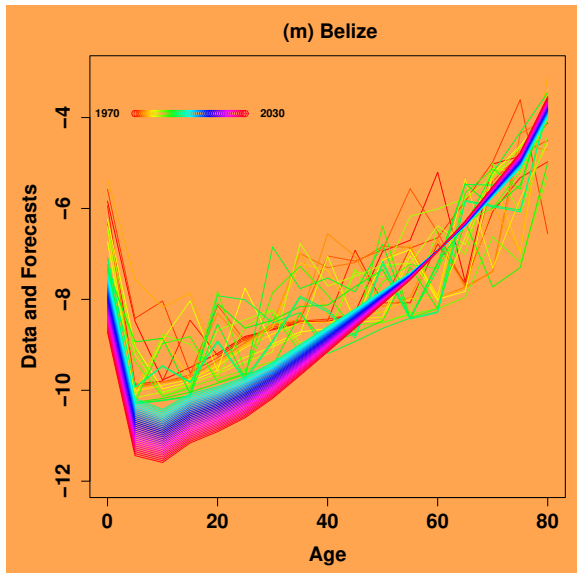
Mortality from Respiratory Infections, males, $\sigma = 0.09$

Smoothing over Age Groups



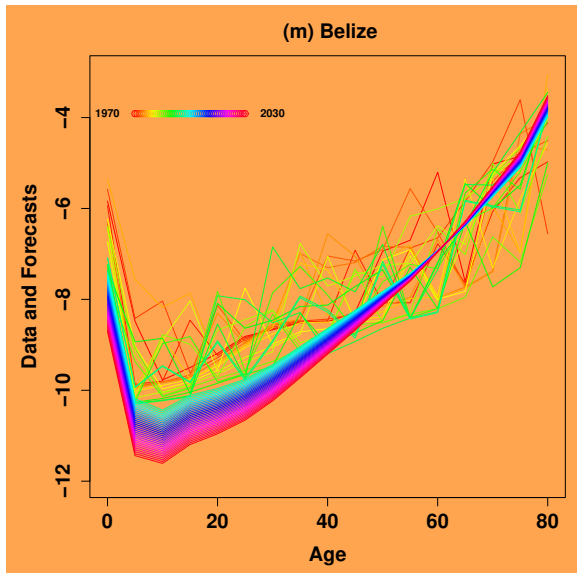
Mortality from Respiratory Infections, males, $\sigma = 0.07$

Smoothing over Age Groups



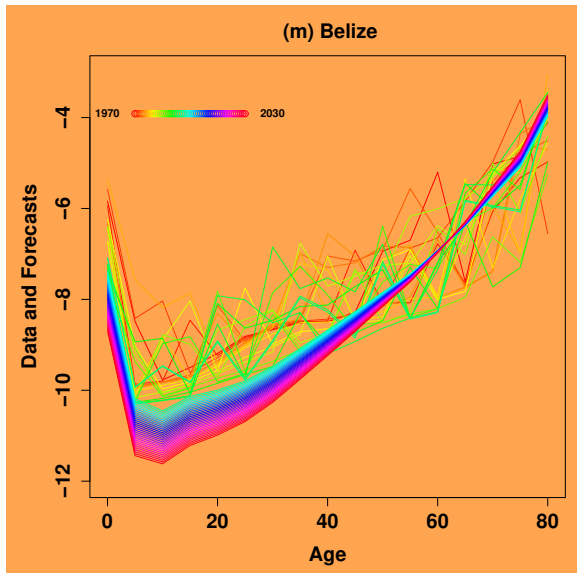
Mortality from Respiratory Infections, males, $\sigma = 0.05$

Smoothing over Age Groups



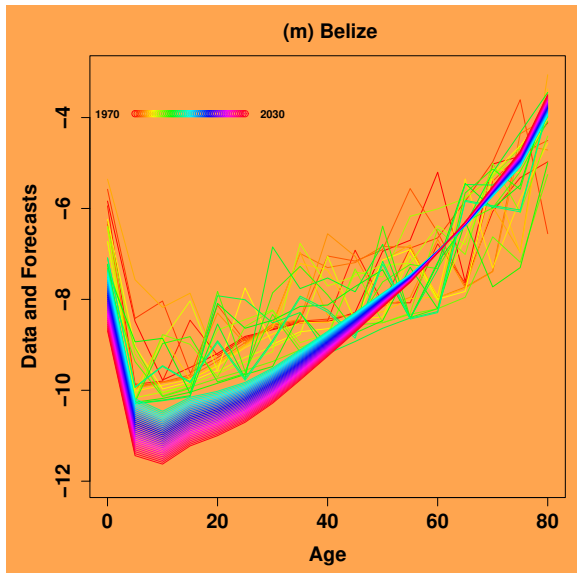
Mortality from Respiratory Infections, males, $\sigma = 0.04$

Smoothing over Age Groups



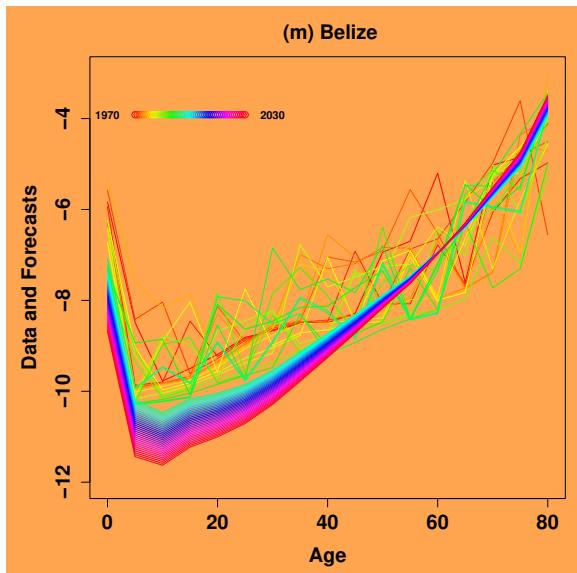
Mortality from Respiratory Infections, males, $\sigma = 0.03$

Smoothing over Age Groups



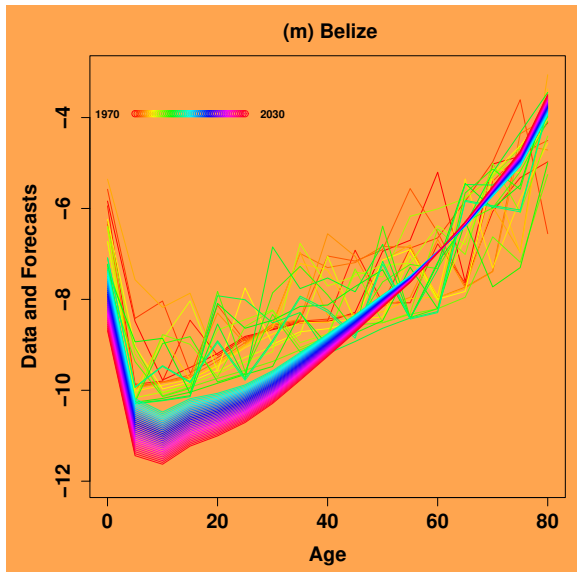
Mortality from Respiratory Infections, males, $\sigma = 0.02$

Smoothing over Age Groups



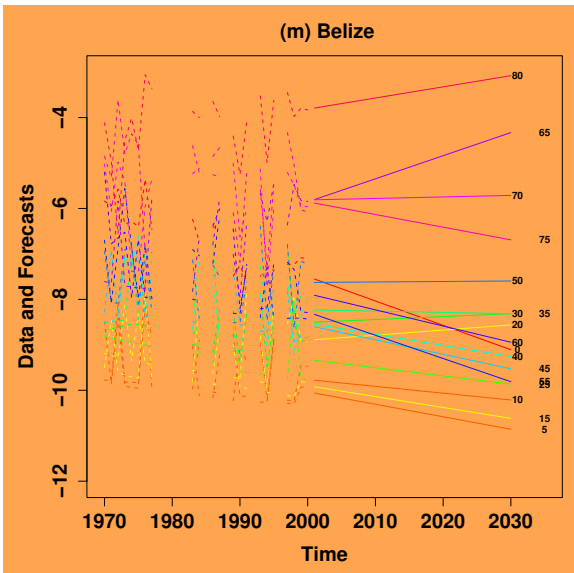
Mortality from Respiratory Infections, males, $\sigma = 0.01$

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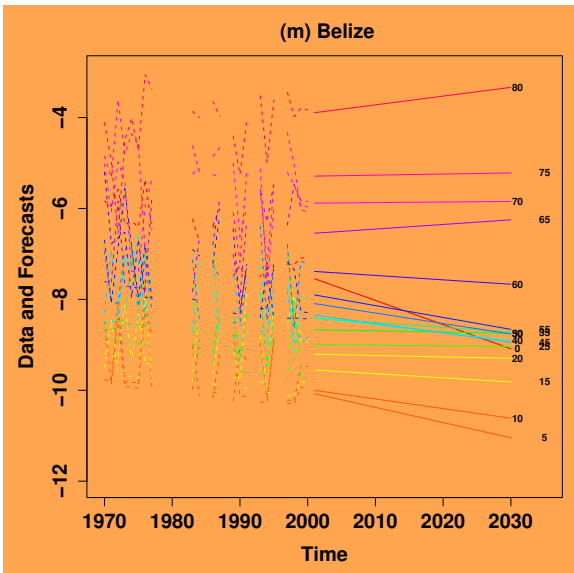
Mortality from Respiratory Infections, males

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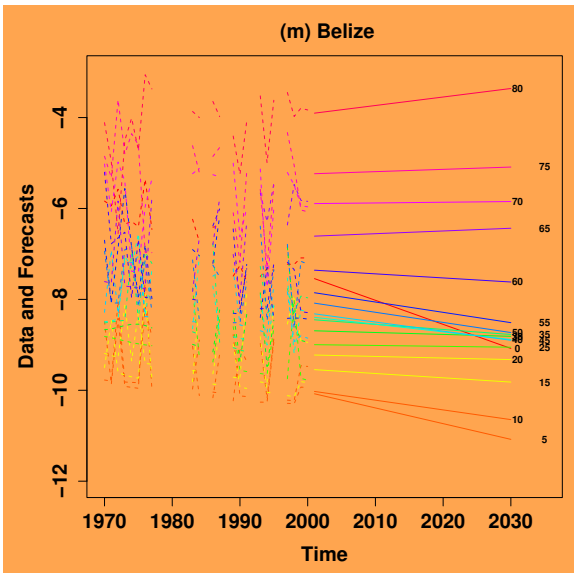
Mortality from Respiratory Infections, males, $\sigma = 2.00$

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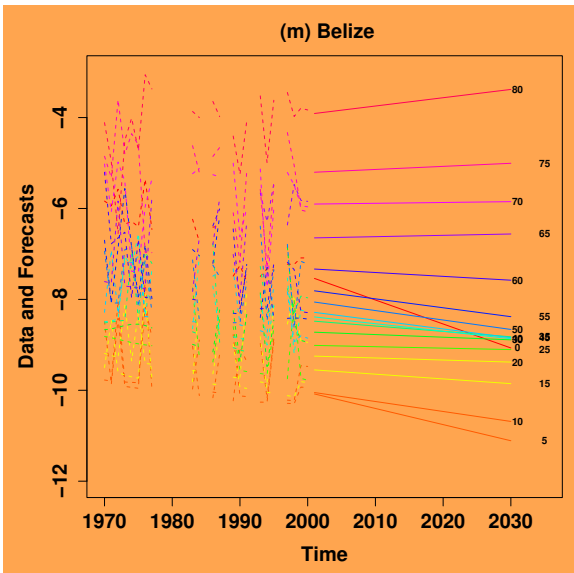
Mortality from Respiratory Infections, males, $\sigma = 1.51$

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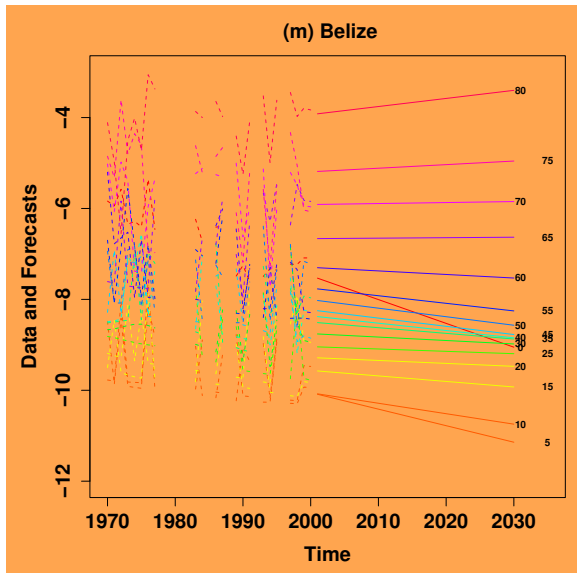
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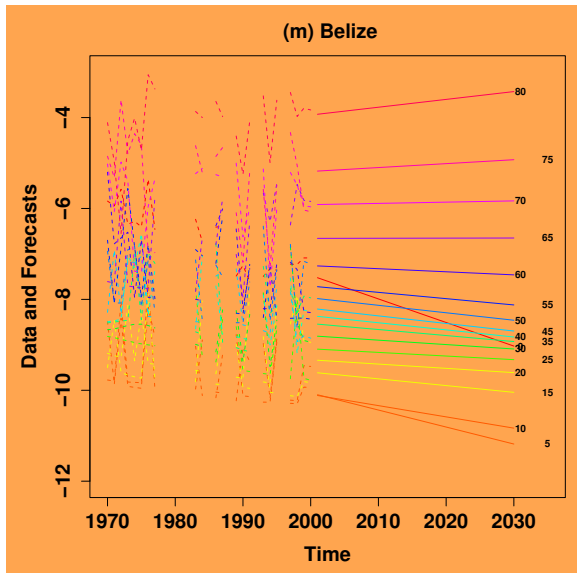
Mortality from Respiratory Infections, males, $\sigma = 0.87$

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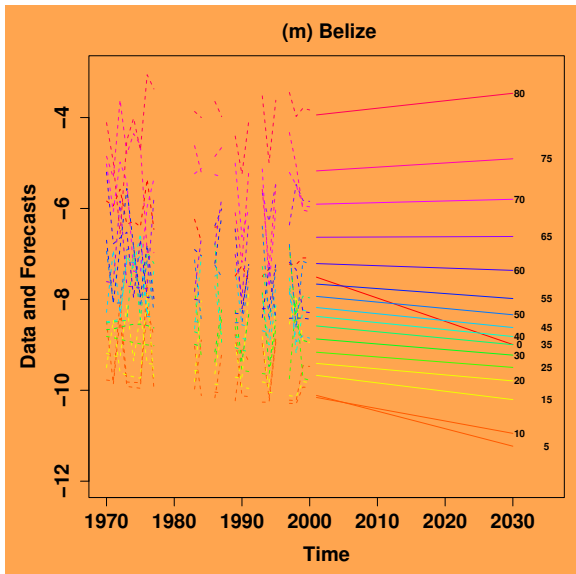
Mortality from Respiratory Infections, males, $\sigma = 0.66$

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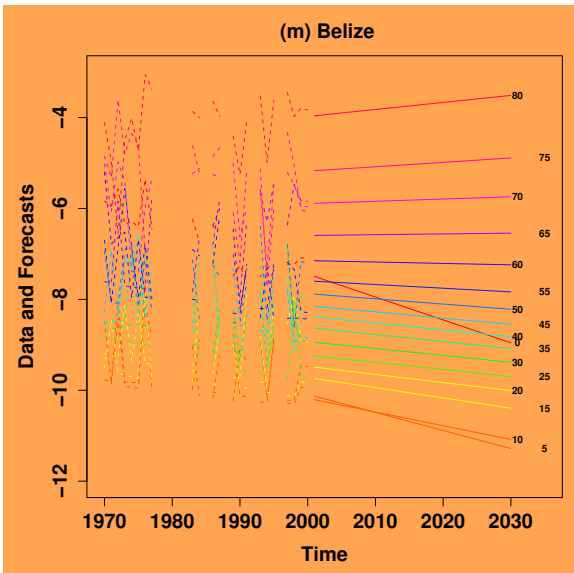
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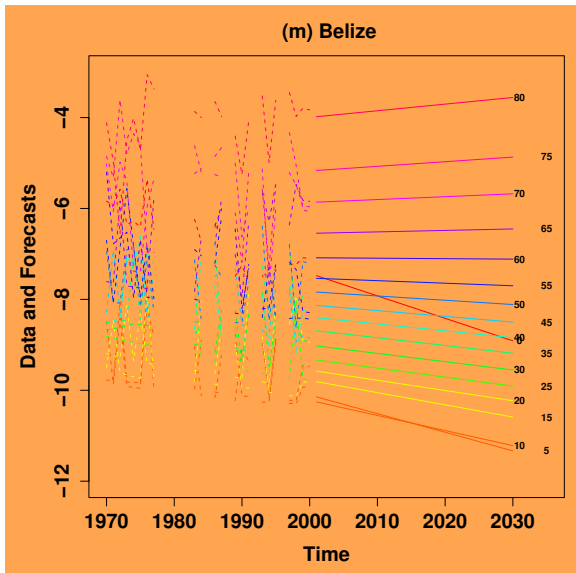
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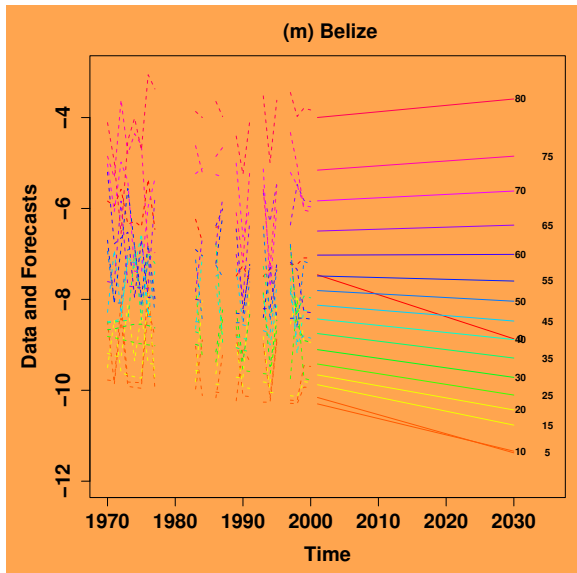
Mortality from Respiratory Infections, males, $\sigma = 0.28$

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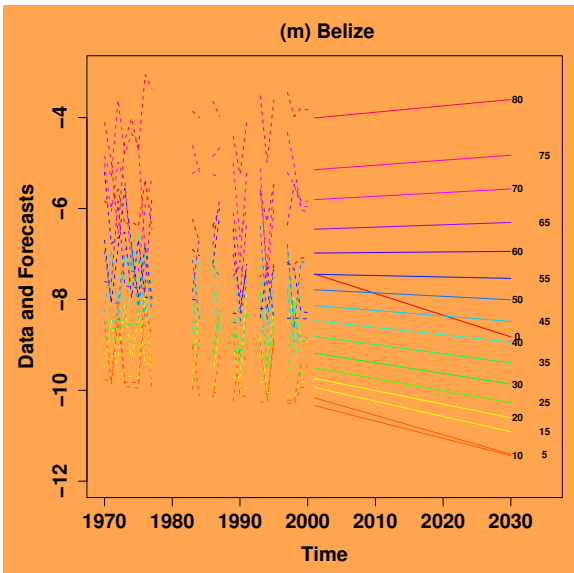
Mortality from Respiratory Infections, males, $\sigma = 0.21$

Smoothing over Age Groups



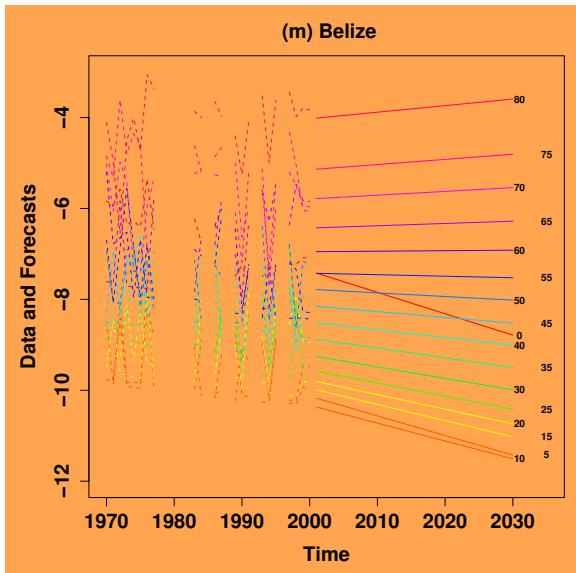
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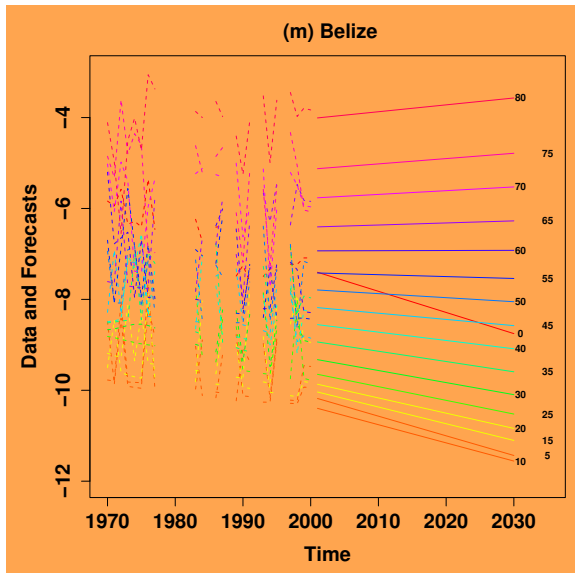
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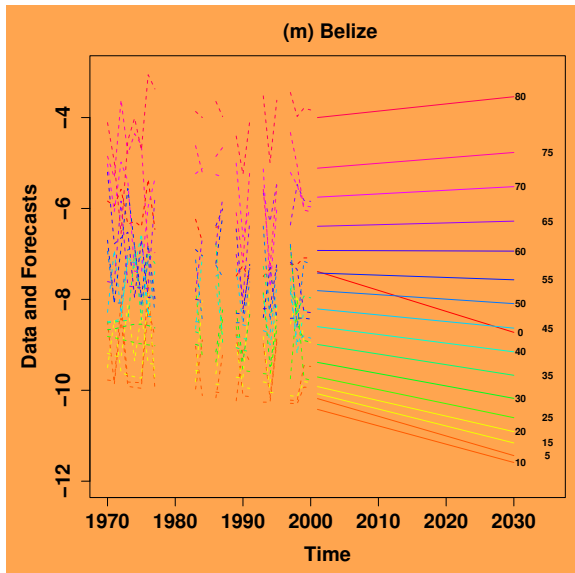
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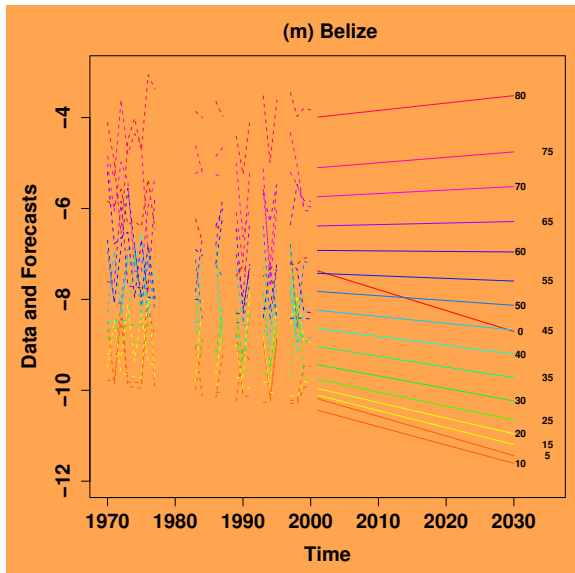
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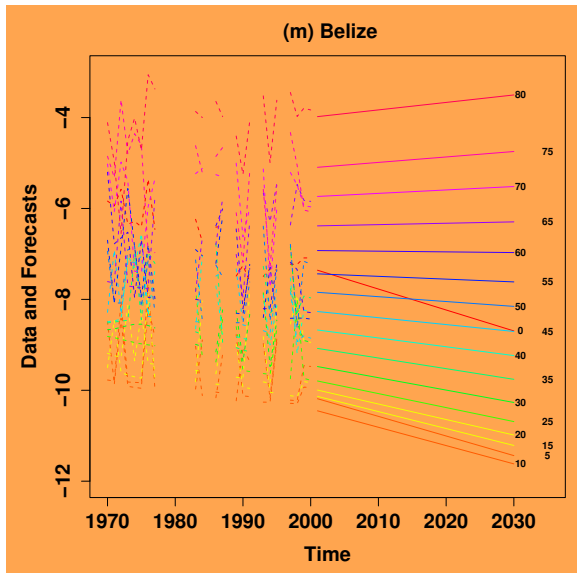
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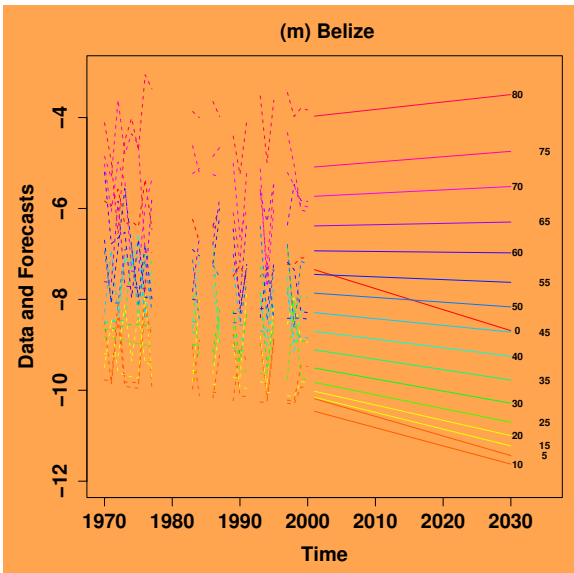
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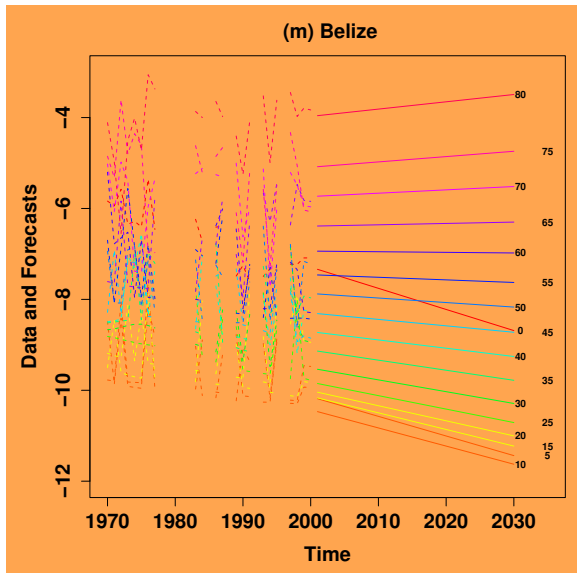
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Smoothing over Age Groups



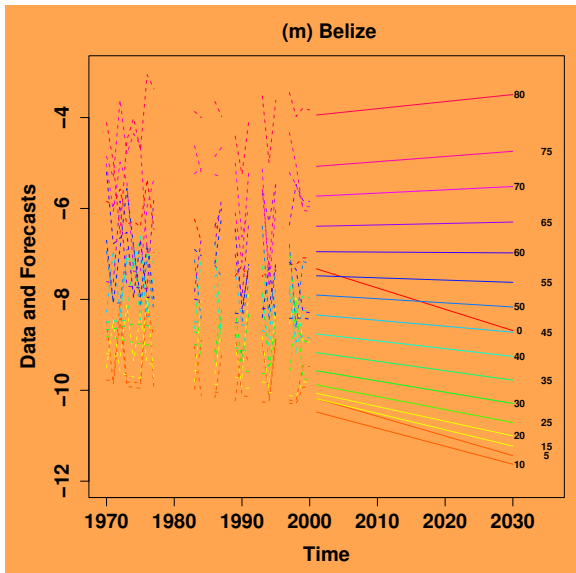
Mortality from Respiratory Infections, males, $\sigma = 0.02$

Smoothing over Age Groups



Mortality from Respiratory Infections, males, $\sigma = 0.01$

Smoothing over Age Groups



Smoothing Trends over Age Groups

Smoothing Trends over Age Groups

Log-mortality in Belize males from respiratory infections

Smoothing Trends over Age Groups

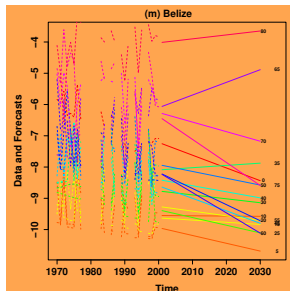
Log-mortality in Belize males from respiratory infections

Least Squares

Smoothing Trends over Age Groups

Log-mortality in Belize males from respiratory infections

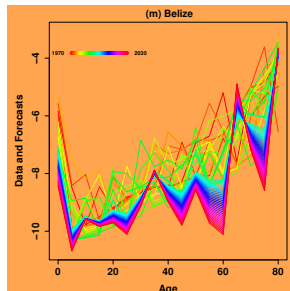
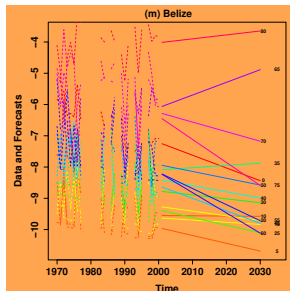
Least Squares



Smoothing Trends over Age Groups

Log-mortality in Belize males from respiratory infections

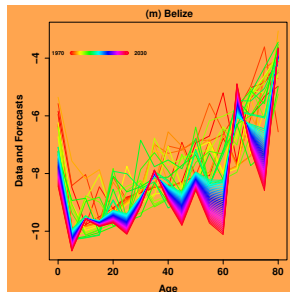
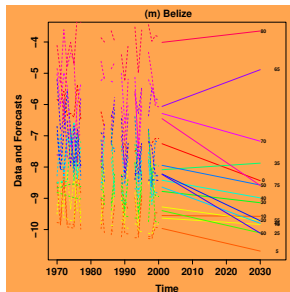
Least Squares



Smoothing Trends over Age Groups

Log-mortality in Belize males from respiratory infections

Least Squares

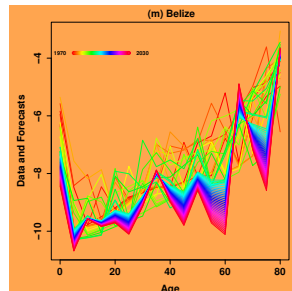
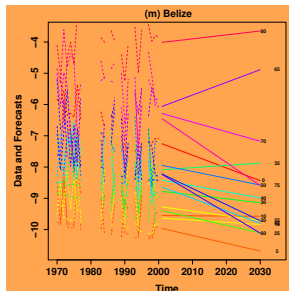


Smoothing
Age Groups

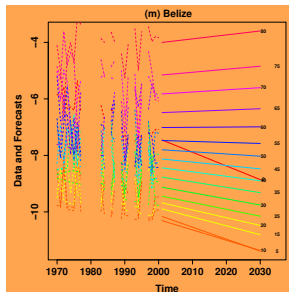
Smoothing Trends over Age Groups

Log-mortality in Belize males from respiratory infections

Least Squares



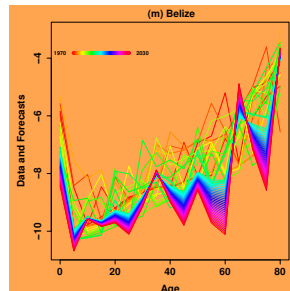
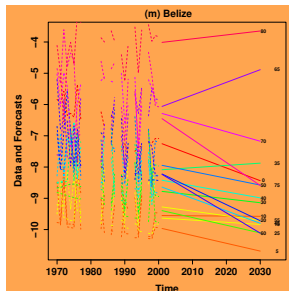
Smoothing
Age Groups



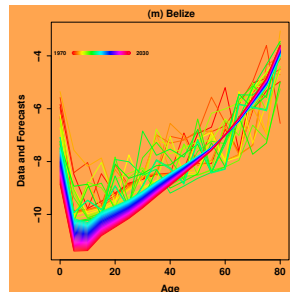
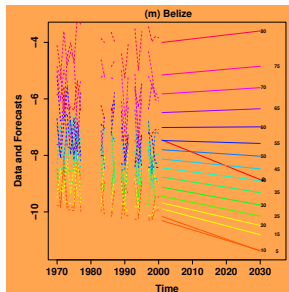
Smoothing Trends over Age Groups

Log-mortality in Belize males from respiratory infections

Least Squares



Smoothing
Age Groups



Smoothing Trends over Age Groups and Time

Smoothing Trends over Age Groups and Time

Log-Mortality in Bulgarian males from respiratory infections

Smoothing Trends over Age Groups and Time

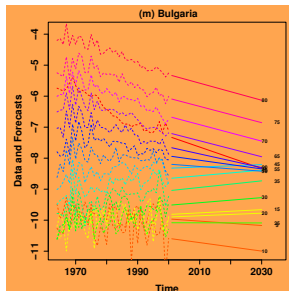
Log-Mortality in Bulgarian males from respiratory infections

Least Squares

Smoothing Trends over Age Groups and Time

Log-Mortality in Bulgarian males from respiratory infections

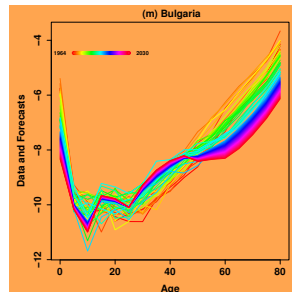
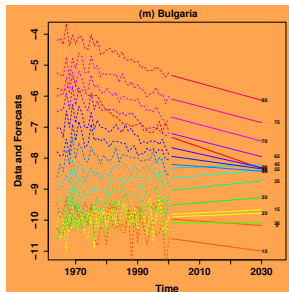
Least Squares



Smoothing Trends over Age Groups and Time

Log-Mortality in Bulgarian males from respiratory infections

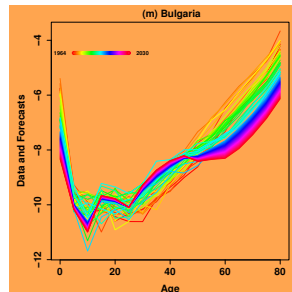
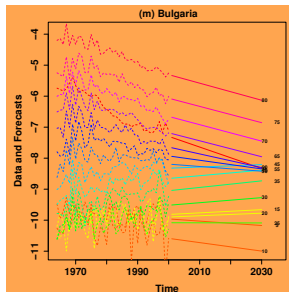
Least Squares



Smoothing Trends over Age Groups and Time

Log-Mortality in Bulgarian males from respiratory infections

Least Squares

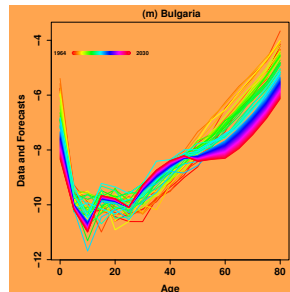
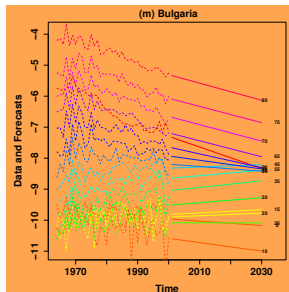


Smoothing
Age and Time

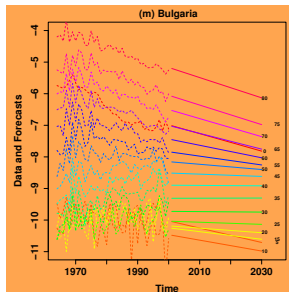
Smoothing Trends over Age Groups and Time

Log-Mortality in Bulgarian males from respiratory infections

Least Squares



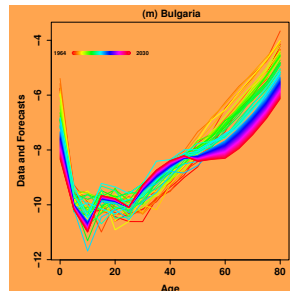
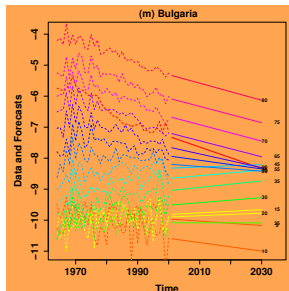
Smoothing
Age and Time



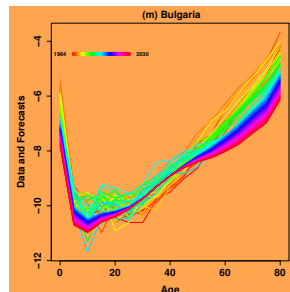
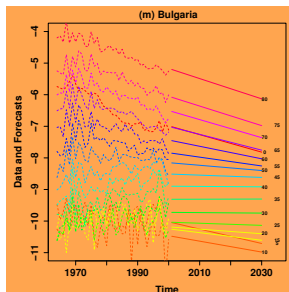
Smoothing Trends over Age Groups and Time

Log-Mortality in Bulgarian males from respiratory infections

Least Squares



Smoothing
Age and Time



Using Covariates (GDP, tobacco, trend, log trend)

Using Covariates (GDP, tobacco, trend, log trend)

Lung cancer in Korean Males

Using Covariates (GDP, tobacco, trend, log trend)

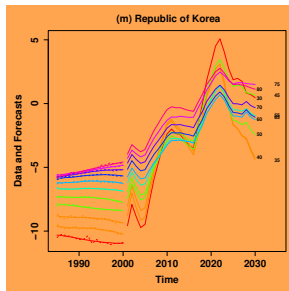
Lung cancer in Korean Males

Least Squares

Using Covariates (GDP, tobacco, trend, log trend)

Lung cancer in Korean Males

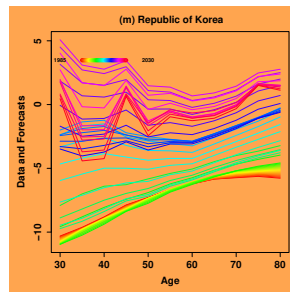
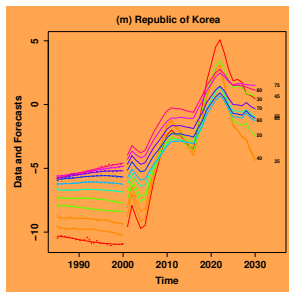
Least Squares



Using Covariates (GDP, tobacco, trend, log trend)

Lung cancer in Korean Males

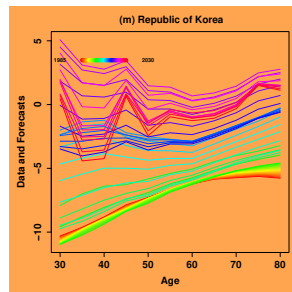
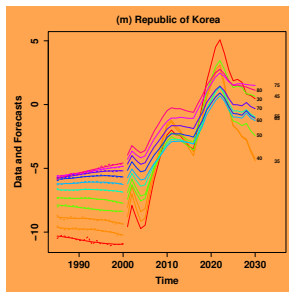
Least Squares



Using Covariates (GDP, tobacco, trend, log trend)

Lung cancer in Korean Males

Least Squares

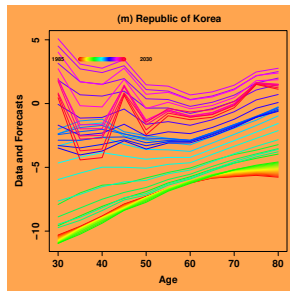
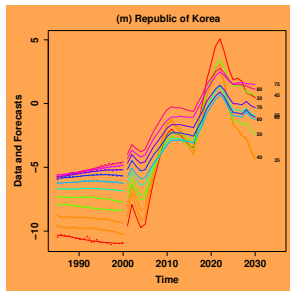


Smooth over age,
time, age/time

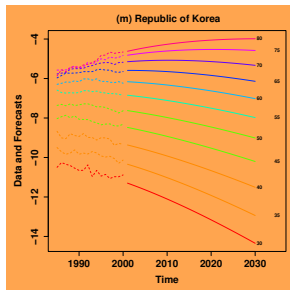
Using Covariates (GDP, tobacco, trend, log trend)

Lung cancer in Korean Males

Least Squares



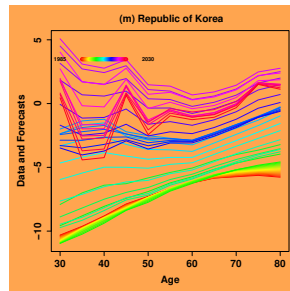
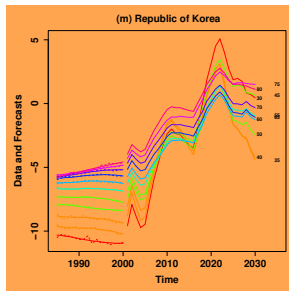
Smooth over age,
time, age/time



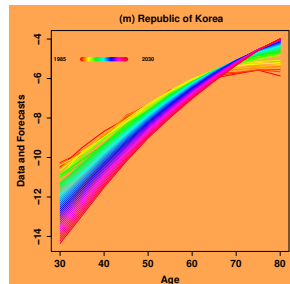
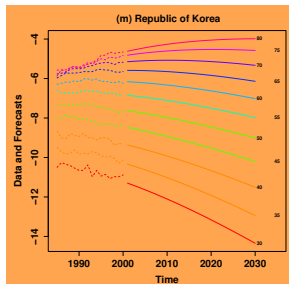
Using Covariates (GDP, tobacco, trend, log trend)

Lung cancer in Korean Males

Least Squares



Smooth over age,
time, age/time



Using Covariates (GDP, tobacco, trend, log trend)

Using Covariates (GDP, tobacco, trend, log trend)

Lung cancer in Males, Singapore

Using Covariates (GDP, tobacco, trend, log trend)

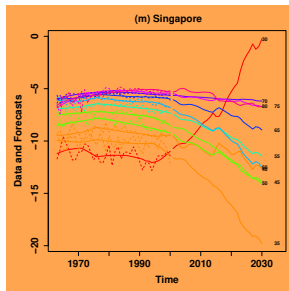
Lung cancer in Males, Singapore

Least Squares

Using Covariates (GDP, tobacco, trend, log trend)

Lung cancer in Males, Singapore

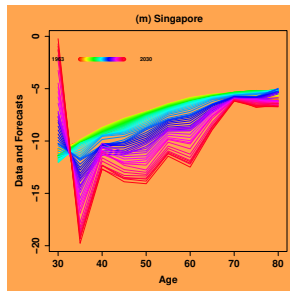
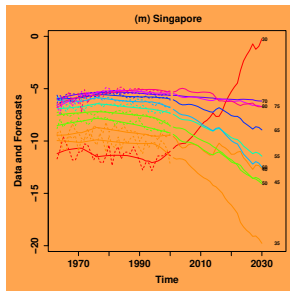
Least Squares



Using Covariates (GDP, tobacco, trend, log trend)

Lung cancer in Males, Singapore

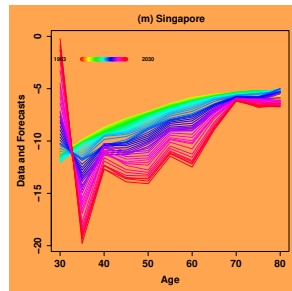
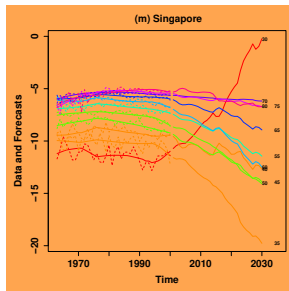
Least Squares



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Least Squares

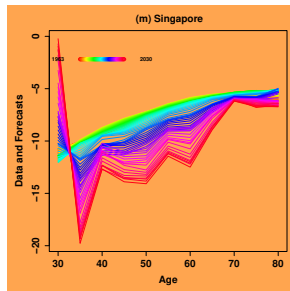
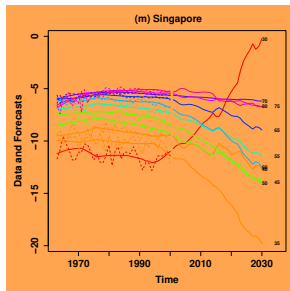


Smooth over age,
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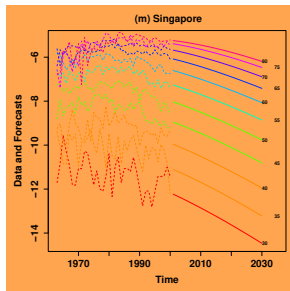
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Lung cancer in Males, Singapore

Least Squares



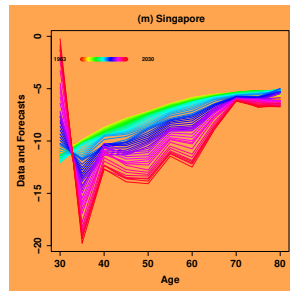
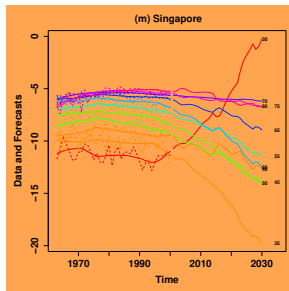
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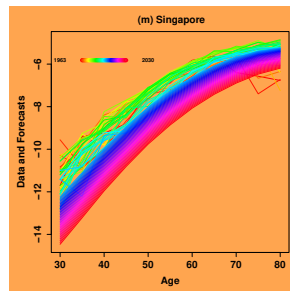
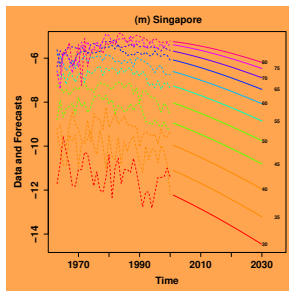
Using Covariates (GDP, tobacco, trend, log trend)

Lung cancer in Males, Singapore

Least Squares

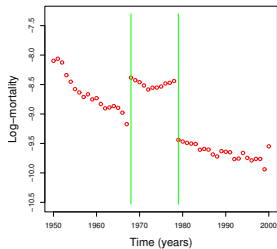


Smooth over age,
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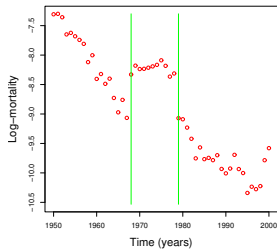


What about ICD Changes?

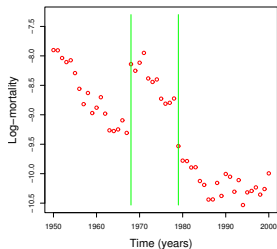
Other Infectious Diseases : USA , age 0 (m)



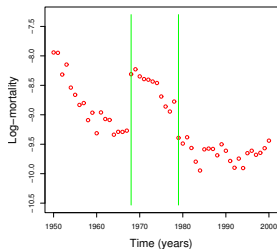
Other Infectious Diseases : France , age 0 (m)



Other Infectious Diseases : Australia , age 0 (m)

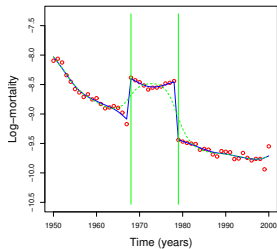


Other Infectious Diseases : United Kingdom , age 0 (m)

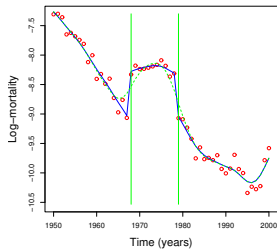


Fixing ICD Changes

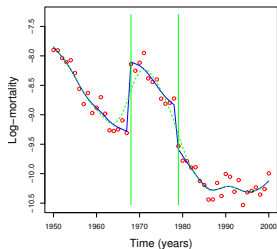
Other Infectious Diseases : USA , age 0 (m)



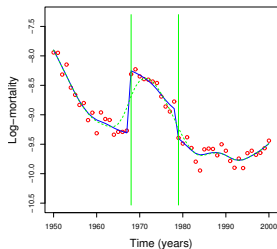
Other Infectious Diseases : France , age 0 (m)



Other Infectious Diseases : Australia , age 0 (m)



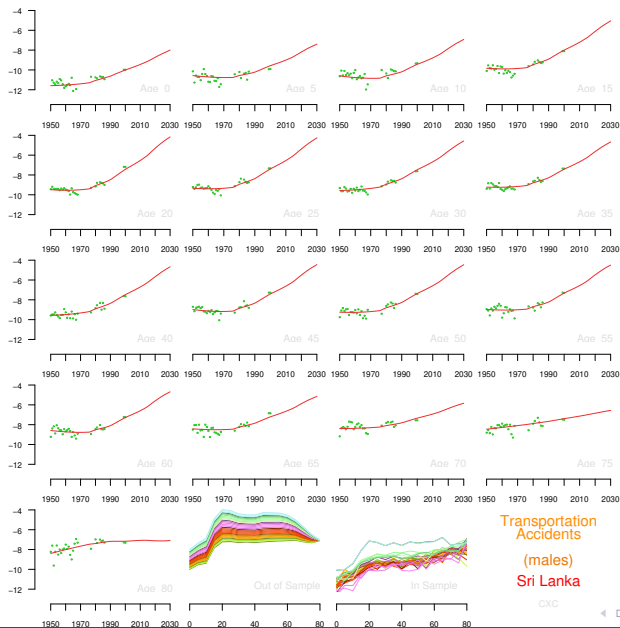
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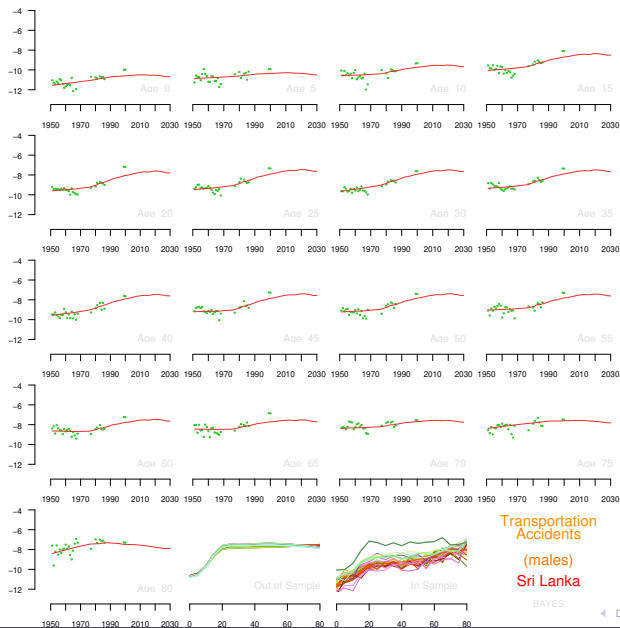
A book manuscript, YourCast software, etc.

<http://GKing.Harvard.edu>

Without Country Smoothing



With Country Smoothing



Formalizing Similarity

Formalizing Similarity

Standard Bayesian Approach

Formalizing Similarity

Standard Bayesian Approach

- Assume **coefficients** are similar

Formalizing Similarity

Standard Bayesian Approach

- Assume **coefficients** are similar
 - But we know little about the coefficients

Formalizing Similarity

Standard Bayesian Approach

- Assume **coefficients** are similar
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- Requires the same covariates in each cross-section

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 - Why measure water quality in the U.S.?

Formalizing Similarity

Standard Bayesian Approach

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 - Why measure water quality in the U.S.?
- Requires covariates with the same meaning in each cross-section

Formalizing Similarity

Standard Bayesian Approach

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Alternative Approach

- Assume **expected mortality** is similar
- Coefficients are unobserved, mortality patterns are well known

Formalizing Similarity

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Alternative Approach

- Assume **expected mortality** is similar
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- Different covariates allowed in each cross-section

Formalizing Similarity

Standard Bayesian Approach

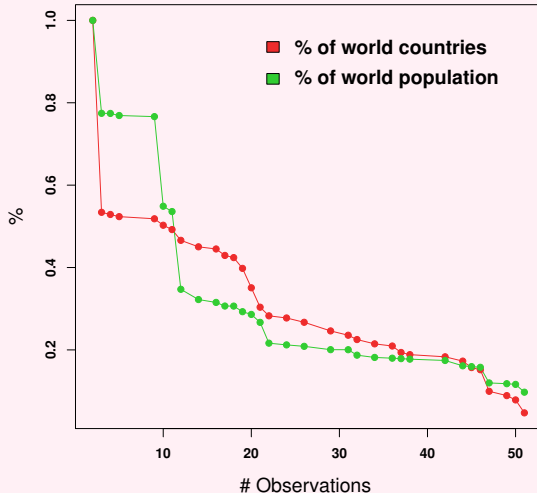
- Assume **coefficients** are similar
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 - Why measure water quality in the U.S.?
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Alternative Approach

- Assume **expected mortality** is similar
- Coefficients are unobserved, mortality patterns are well known
- Different covariates allowed in each cross-section
- Covariates with the same name can have different meanings

Many Short Time Series

Coverage of WHO data base (age specific, all causes)



Preview of Results: Out-of-Sample Evaluation

Preview of Results: Out-of-Sample Evaluation

Mean Absolute Error in Males (over age and country)

Preview of Results: Out-of-Sample Evaluation

Mean Absolute Error in Males (over age and country)

	Mean Absolute Error			% Improvement	
	Best Previous	Our Method	Best Conceivable	Over Best Previous	to Best Conceivable
Cardiovascular	0.34	0.27	0.19	22	49
Lung Cancer	0.36	0.27	0.17	24	47
Transportation	0.37	0.31	0.18	16	31
Respiratory Chronic	0.45	0.39	0.26	13	30
Other Infectious	0.55	0.48	0.32	12	30
Stomach Cancer	0.30	0.27	0.20	8	24
All-Cause	0.17	0.15	0.08	12	22
Suicide	0.31	0.29	0.18	7	17
Respiratory Infectious	0.49	0.47	0.28	3	7

Preview of Results: Out-of-Sample Evaluation

Mean Absolute Error in Males (over age and country)

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Stomach Cancer	0.30	0.27	0.20	8	24
All-Cause	0.17	0.15	0.08	12	22
Suicide	0.31	0.29	0.18	7	17
Respiratory Infectious	0.49	0.47	0.28	3	7

- Each row averages 6,800 forecast errors (17 age groups, 40 countries, and 10 out-of-sample years).

Preview of Results: Out-of-Sample Evaluation

Mean Absolute Error in Males (over age and country)

	Mean Absolute Error			% Improvement	
	Best Previous	Our Method	Best Conceivable	Over Best Previous	to Best Conceivable
Cardiovascular	0.34	0.27	0.19	22	49
Lung Cancer	0.36	0.27	0.17	24	47
Transportation	0.37	0.31	0.18	16	31
Respiratory Chronic	0.45	0.39	0.26	13	30
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- The new method out-performs with the same covariates, for most countries, causes, sexes, and age groups.
- Does much better with better covariates